Polytechnic University of Bari

PIXEL AVATAR DDPM

Tiny Text-Conditioned Avatar Diffusion from Scratch

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RESEARCH QUESTION

“Can a very small diffusion model, whose denoiser and text encoder are both trained from scratch, generalize to combinations of avatar attributes that were not observed together during training?”

DATASET



Name	Google Cartoon Set
Samples	10,000
Resolution	500 x 500
Artworks	10 Attributes, 171 Variants: chin_length, eye_angle, eye_lashes, eye_lid, eyebrow_shape, eyebrow_weight, face_shape, facial_hair, glasses,hair
Colors	4 Attributes, 33 Variants: eye_color, face_color, glasses_color, hair_color
Proportions	4 Attributes, 13 Variants: eye_eyebrow_distance, eye_slant, eyebrow_thickness, eyebrow_width

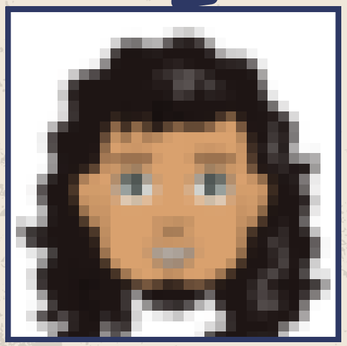
PREPROCESSING



500

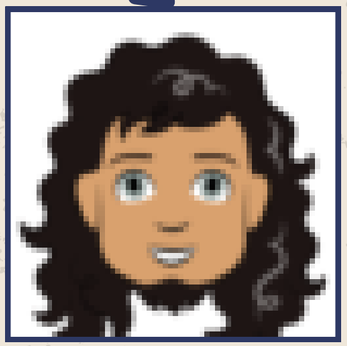
500

CROPPING + DOWNSAMPLING



32

32



64

64

```
eye_angle," 1"," 3"
eye_lashes," 1"," 2"
eye_lid," 1"," 2"
chin_length," 1"," 3"
eyebrow_weight," 1"," 2"
eyebrow_shape," 9"," 14"
eyebrow_thickness," 0"," 4"
face_shape," 4"," 7"
facial_hair," 4"," 15"
hair," 25"," 111"
eye_color," 3"," 5"
face_color," 4"," 11"
hair_color," 7"," 10"
glasses," 11"," 12"
glasses_color," 4"," 7"
eye_slant," 2"," 3"
eyebrow_width," 1"," 3"
eye_eyebrow_distance," 0"," 3"
```

MAPPING + CAPTION GENERATION

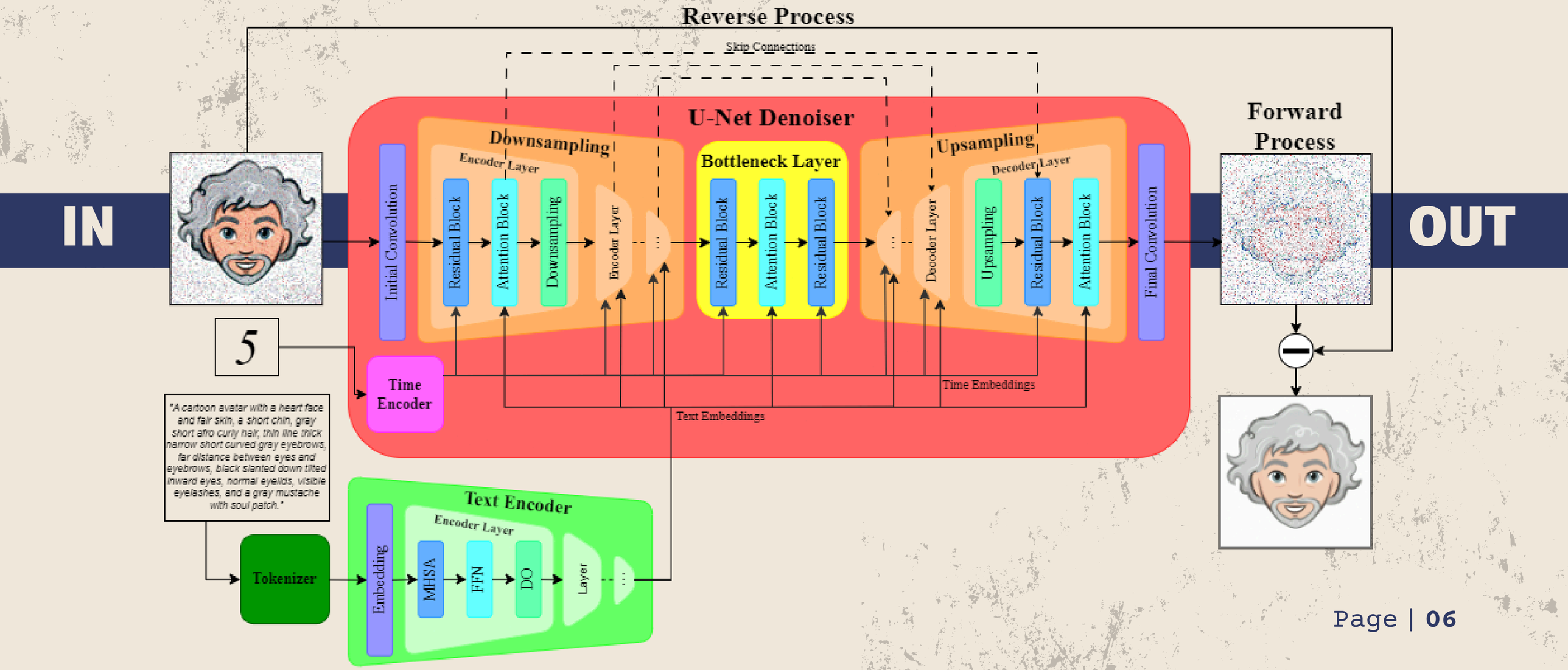
“A cartoon avatar with a heart face and light tan skin, a medium length chin, black shoulder length wavy hair, thick line thin medium width rounded thin black eyebrows, medium distance between eyes and eyebrows, gray slanted down eyes, single eyelids, visible eyelashes, and a black goatee.”

COMPOSITIONAL
SPLITS

Combination #1	hair_color: *“black”* facial_hair: *“mustache with goatee”*	33 Samples
Combination #2	hair_color: *“red”* eye_color: *“blue”*	203 Samples
Combination #3	face_color: *“fair pinkish”* hair: *“top knot bun”*	12 Samples
Combination #4	glasses: *“cat eye dark sunglasses”* hair_color: *“yellow”*	47 Samples



METHODOLOGY



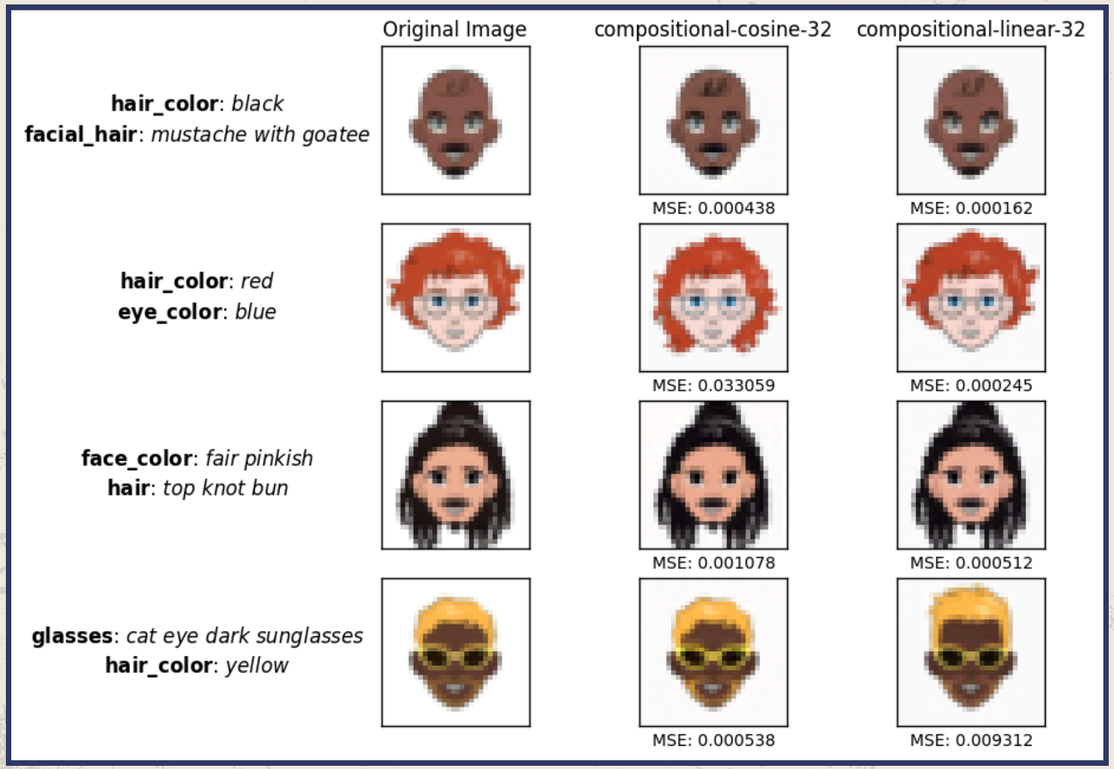
CONFIGURATIONS

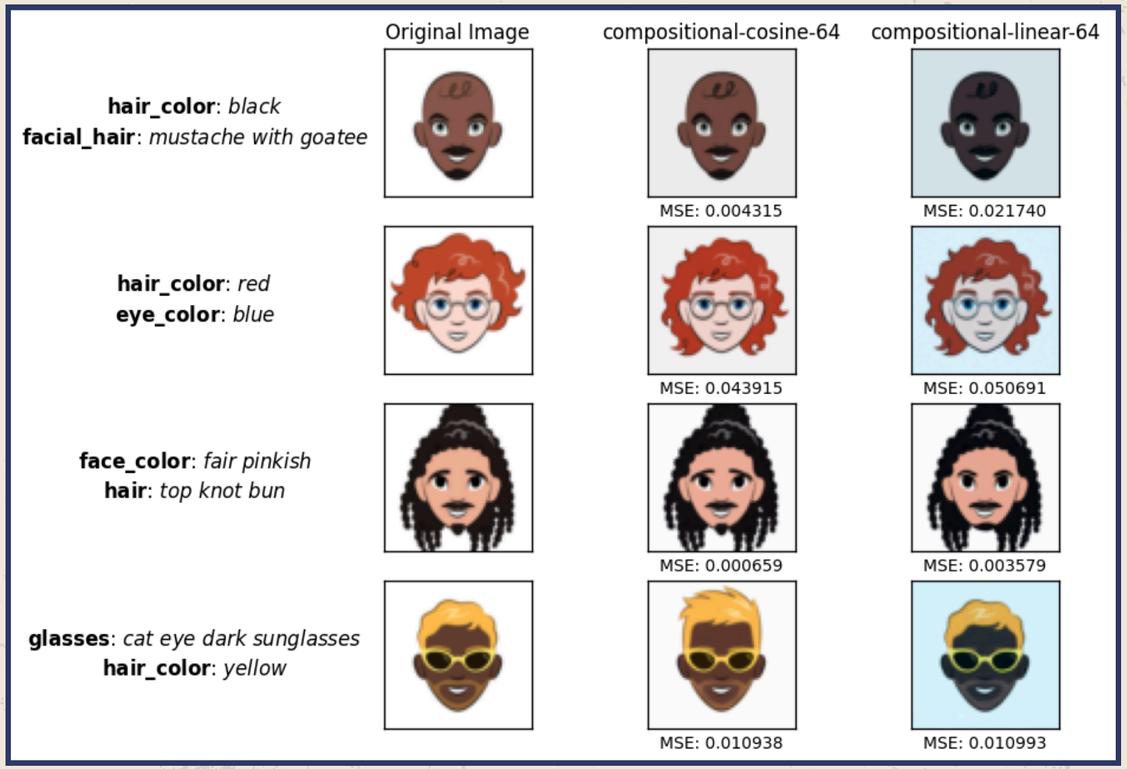
Model Configuration	$H \times W$	C_0	L	A	D_{time}	S	E	D_{text}	DO	Epochs	B	η_0	ω	Schedule Strategy
unconditioned-32	32×32	64	3	4	256	70	4	128	0.1	50	32	$2 \cdot 10^{-4}$	10^{-2}	Scaled Linear
conditioned-ordinary-32	32×32	64	3	4	256	70	4	128	0.1	50	32	$2 \cdot 10^{-4}$	10^{-2}	Scaled Linear
compositional-linear-32	32×32	**128**	3	4	**512**	70	4	128	0.1	50	32	$2 \cdot 10^{-4}$	10^{-2}	Scaled Linear
compositional-cosine-32	32×32	64	3	4	128	70	4	128	0.1	50	32	$2 \cdot 10^{-4}$	10^{-2}	Cosine
unconditioned-64	64×64	128	3	4	512	70	4	128	0.1	25	10	$2 \cdot 10^{-4}$	10^{-2}	Scaled Linear
conditioned-ordinary-64	64×64	128	3	4	512	70	4	128	0.1	25	10	$2 \cdot 10^{-4}$	10^{-2}	Scaled Linear
compositional-linear-64	64×64	**64**	3	4	**256**	70	4	128	0.1	25	10	$2 \cdot 10^{-4}$	10^{-2}	Scaled Linear
compositional-cosine-64	64×64	128	3	4	512	70	4	128	0.1	25	10	$2 \cdot 10^{-4}$	10^{-2}	Cosine

Model Configuration	**Evaluated Split**	**Parameters**	**Size** (MB)	**FID**	**KID** ($\times 10^{-3}$)	**Diversity Score** ($\times 10^{-3}$)
unconditioned-32	ordinary_32	37,782,787	144.12	50.8532	3.738 ± 4.011	82.020
conditioned-ordinary-32	ordinary_32	37,782,787	144.12	52.3479	1.481 ± 2.487	0.047
compositional-linear-32	compositional_32	147,603,459	563.06	23.7674	-1.480 ± 2.147	0.037
compositional-cosine-32	compositional_32	37,782,787	144.12	27.9363	-1.029 ± 2.669	6.267
unconditioned-64	ordinary_64	147,603,331	563.06	31.3289	3.216 ± 2.289	96.902
conditioned-ordinary-64	ordinary_64	147,603,331	563.06	28.9332	0.867 ± 1.814	3.410
compositional-linear-64	compositional_64	37,782,659	144.12	21.0904	1.233 ± 1.740	6.761
compositional-cosine-64	compositional_64	147,603,331	563.06	14.1279	-0.931 ± 1.489	0.181

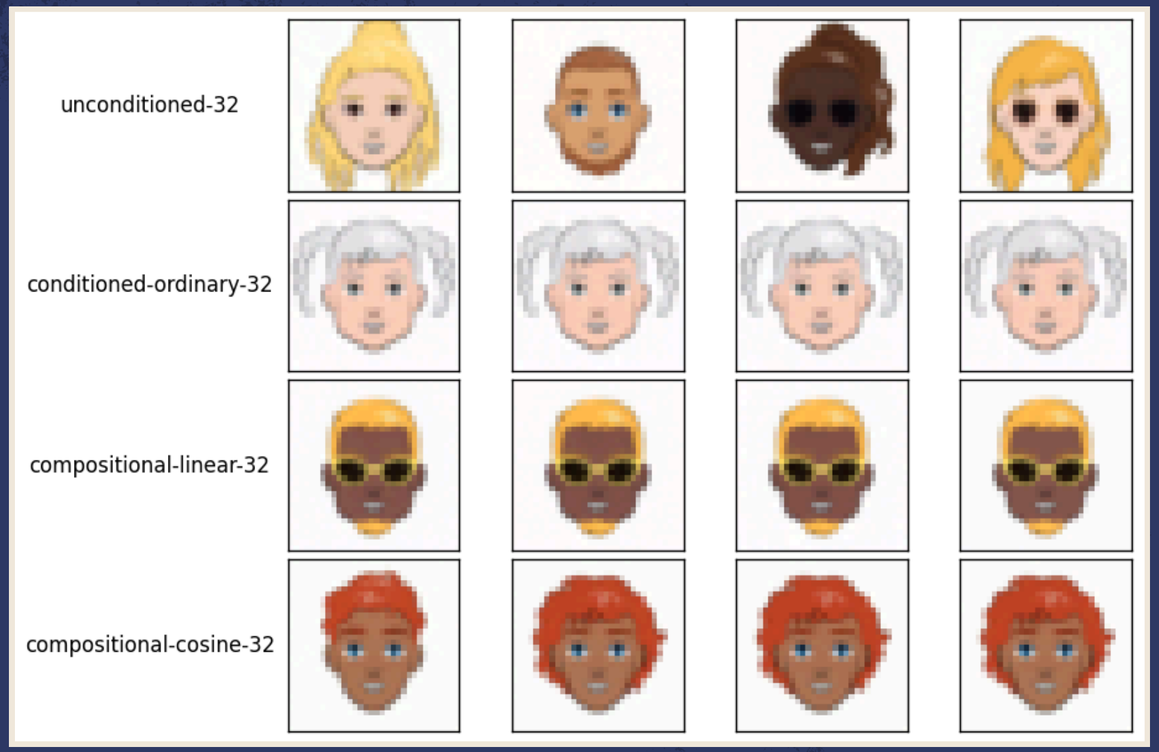
RESULTS

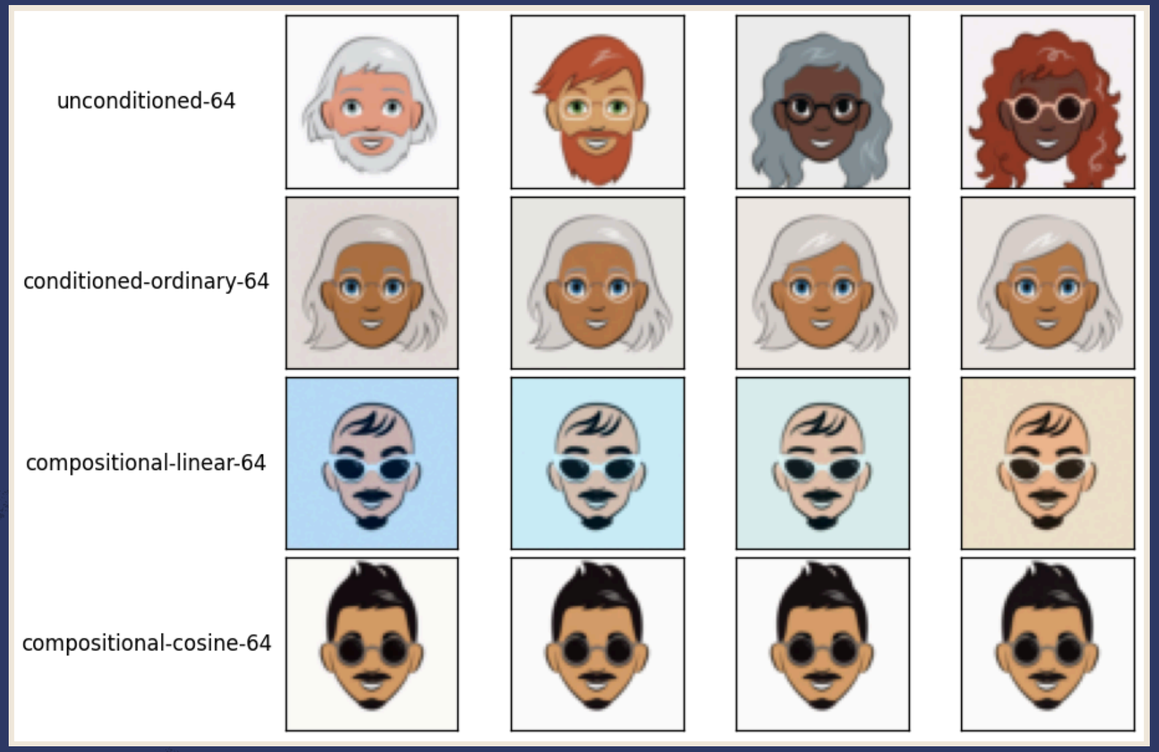
COMPOSITIONAL
GRIDS INSPECTION





DIVERSITY
GRIDS INSPECTION



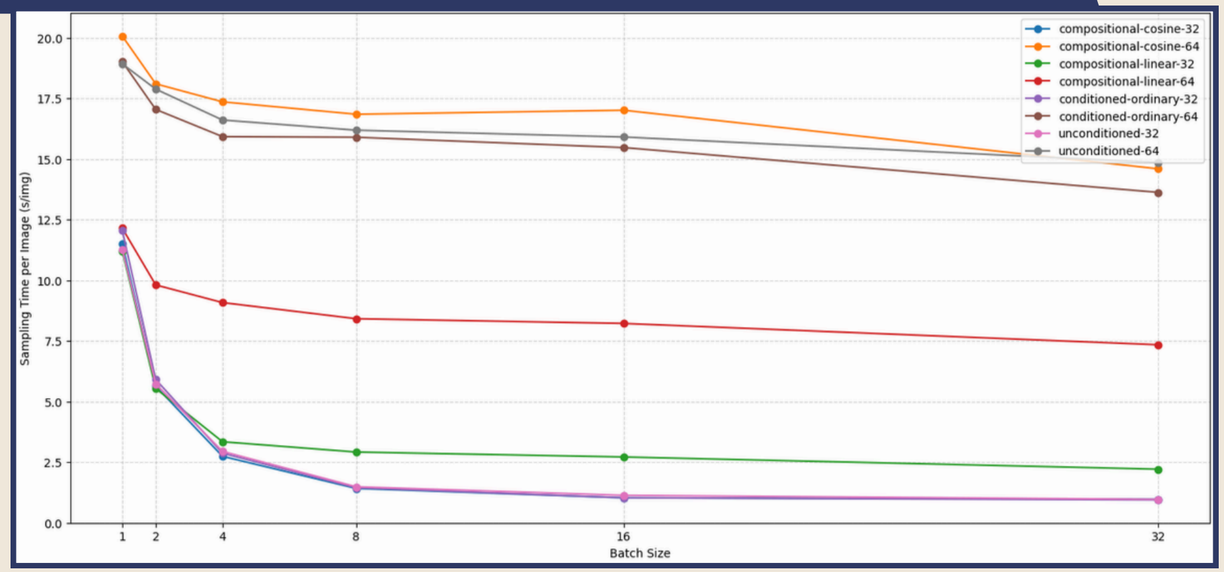


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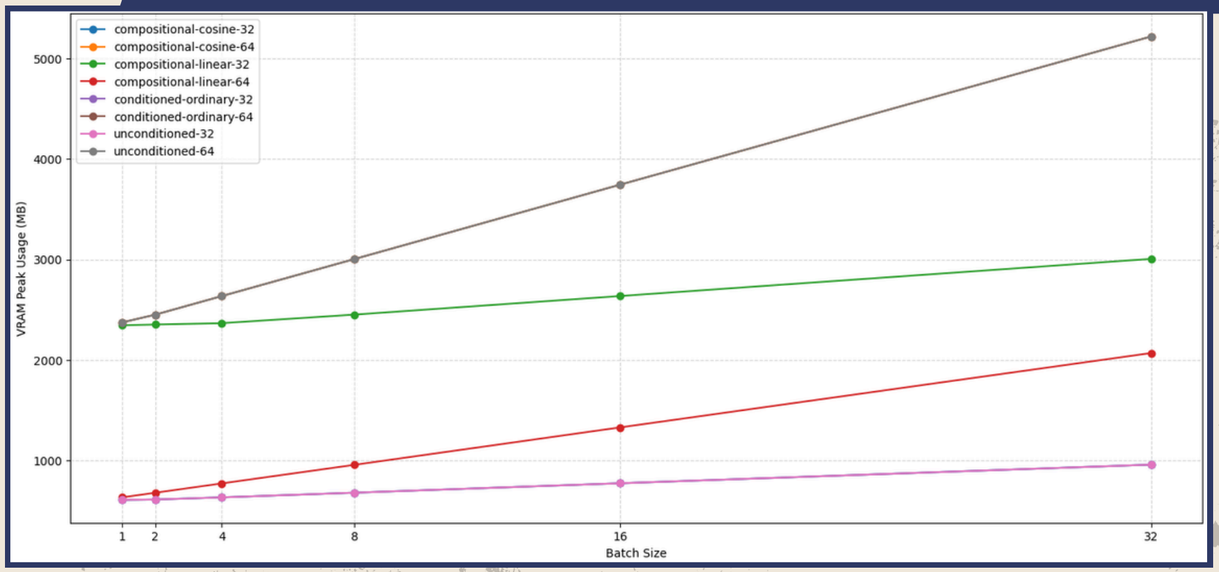
COMPUTATIONAL METRICS

Batch Size:	1	2	4	8	16	32
Model Configuration	**Sampling Time** (s/img)					
unconditioned-32	11.2781	5.7298	2.9500	1.4926	1.1407	0.9736
conditioned-ordinary-32	12.0549	5.9134	2.8751	1.4546	1.0380	0.9489
compositional-linear-32	11.3121	5.6786	3.2869	2.8429	2.6434	2.1889
compositional-cosine-32	11.4941	5.6370	2.7467	1.4233	1.0368	0.9772
unconditioned-64	18.9257	17.8852	16.6111	16.1922	15.9116	14.8583
conditioned-ordinary-64	19.0143	17.0479	15.9268	15.9014	15.4755	13.6288
compositional-linear-64	12.1489	9.8134	9.0851	8.4202	8.2257	7.3467
compositional-cosine-64	18.8560	16.7454	15.6279	15.8473	15.6405	14.4413
Model Configuration	**Peak VRAM Usage** (MB)					
unconditioned-32	604.20	608.01	629.29	675.87	769.93	955.29
conditioned-ordinary-32	604.20	608.01	629.29	675.87	769.93	955.29
compositional-linear-32	2344.41	2351.46	2365.26	2450.59	2635.76	3006.06
compositional-cosine-32	604.20	608.01	629.29	675.87	769.93	955.29
unconditioned-64	2372.90	2450.14	2634.87	3004.31	3745.17	5222.12
conditioned-ordinary-64	2372.90	2450.14	2634.87	3004.31	3745.17	5222.12
compositional-linear-64	629.44	675.43	768.15	953.59	1326.38	2067.54
compositional-cosine-64	2372.90	2450.14	2634.87	3004.31	3745.17	5222.12

SAMPLING TIME



VRAM PEAK USAGE



CONCLUSIONS

Research Answer: A tiny pixel-space DDPM, with denoiser and text encoder trained from scratch, generates attribute combinations never seen together in training.

Best results: cosine schedule at 64×64 (FID 14.13); at 32×32 a small model is enough.

Limitations: low diversity across seeds (overfitting, high CFG scale) and color/background artifacts at 64×64.

Future work: lower CFG scale, EMA, larger dataset (100k), DDIM / Flow Matching.

THANK YOU!